

Muons cosmiques

Session 4

TP Césire

2018/2019

LPSC, Grenoble

by Kseniia Svirina

Result of our measurements:

$$v = \frac{d_1^{\text{mean}} - d_2^{\text{mean}}}{|t_1 - t_2|}$$

Uncertainties of the speed of the muon

$$\frac{\Delta v}{v} = \sqrt{\frac{\Delta d_1^2}{(d_1^{\text{mean}})^2} + \frac{\Delta d_2^2}{(d_2^{\text{mean}})^2} + \frac{\Delta t_1^2}{t_1^2} + \frac{\Delta t_2^2}{t_2^2}}$$

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$$\Delta d_1, \Delta d_2$$

***due to the measurements of the
detectors width, length, etc.***

$$\Delta x = 1\text{cm}$$

$$\Delta y = 15\text{cm} \quad \text{-- due to the assumption of
the rectangular shape...}$$

$$\Delta h = 1\text{cm}$$

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Uncertainties from the calculation of the time

$$t_{1,2} = t_{1,2}^{\text{calculated}} \pm \Delta t_{1,2}^{\text{gaussian-fit}} \pm \Delta t_{1,2}^{\text{calibration-TAC}}$$

we can neglect it

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Take σ from your code

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad \Delta x_{1,2} = \frac{\sigma_{1,2}}{\sqrt{N_{1,2}^{\mu \text{ detected}}}}$$

Use the result of the TAC calibration

$$\Delta t_{1,2}^{\text{gaussian-fit}} = b \cdot \Delta x_{1,2} + c$$

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Calculate $\Delta v = \sqrt{\dots} \cdot v$

Write the result for the muon speed in this form:

$$v_{\mu} = (v \pm \Delta v) \cdot 10^8 (\text{m/s})$$

Plan of the report (example)

- **Title page** (*“TP Cesire: Muons Cosmiques”, your names, encadrant name, date...*)
- **Introduction** (*what is a muon, from where they come, what are the cosmic rays, what happens in the atmosphere, lifetime, how can the muons reach the ground-level, ...*)
- **Experiment description** (*about scintillation detectors, PM's, discriminator, coincidence scheme, TAC, all the idea why we make 2 acquisitions, what do we measure...*)
- **Calculation of the speed of the muons** (*simulation of the d-mean, extraction of the Δt from the results of the acquisition, uncertainties, result*)
- **Conclusions** (*Briefly: what did we do, what have we obtained? Is our result physically adequate, if not – why?...*)
- **References** (*if you used any literature*)
- **Annexe** (*codes, pictures, whatever*)

(It'd be also nice to have the main pictures and diagrams in the text)